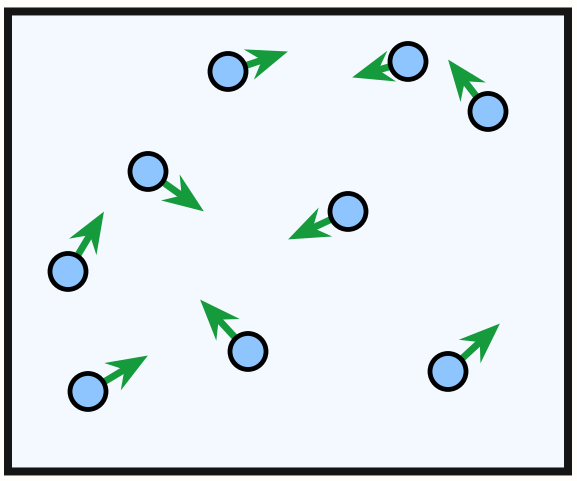
Kinetic Theory of Gases

Class 11 · IDEAL gas in THERMAL EQUILIBRIUM · T in KELVIN · n = MOLES, $n_V = N/V$ = number DENSITY

1 The molecular picture



gas in a box, N molecules

- A gas = a very large number N of molecules in RANDOM motion, far apart compared with their size.
- Pressure = momentum delivered to the walls per unit area per second; temperature measures the average KE of RANDOM motion.
- Macroscopic P, V, T follow from averages over molecules.

2 Symbols used on every page

$$n = \frac{N}{N_A} = \frac{m_{gas}}{M} \qquad n_V = \frac{N}{V} \qquad \rho = n_V m$$

n = number of MOLES · N = number of molecules · n_V = number DENSITY ($\mathrm{m^{-3}}$) · m = mass of ONE molecule (kg) · m_{gas} = mass of the sample · M = MOLAR mass ($\mathrm{kg\,mol^{-1}}$) · ρ = density ($\mathrm{kg\,m^{-3}}$)

- $k = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$ (Boltzmann) · $R = 8.314\,\mathrm{J\,mol^{-1}\,K^{-1}}$ · $N_A = 6.022 \times 10^{23}\,\mathrm{mol^{-1}}$ · $k = R/N_A$, $M = N_A m$.
- Use M in $\mathrm{kg\,mol^{-1}}$ in every speed formula: oxygen $M = 0.032$, not 32.

Syllabus Checklist

JEE Main + JEE Advanced + NEET syllabus + standard results

1 JEE Main + JEE Advanced + NEET syllabus

- ✓ Assumptions of the kinetic theory (ideal gas) — p3
- ✓ Ideal gas equation $PV = nRT = NkT$; gas laws — p4, p5
- ✓ Work done on compressing a gas (quasi-static) — p5
- ✓ Pressure of an ideal gas: derivation, $P = \frac{1}{3}\rho v_{rms}^2$ — p6, p7
- ✓ Kinetic interpretation of temperature, $\frac{3}{2}kT$ — p8
- ✓ rms, mean, most probable speeds and ratios — p9
- ✓ Maxwell speed distribution (qualitative, graph) — p10–p12
- ✓ Degrees of freedom; equipartition of energy — p13, p14
- ✓ Internal energy $U = \frac{f}{2}nRT$ — p14
- ✓ C_V, C_P, γ, Mayer relation; gases, solids — p15, p16
- ✓ Mean free path — p18

2 Also covered: laws + standard results

- ✓ γ and C_V of a gas MIXTURE — p17
- ✓ Dalton's law of partial pressures; Graham's law — p19
- ✓ Real gases: Z vs P, van der Waals (qualitative) — p20
- ✓ C_V vs T (vibration frozen at moderate T) — p16
- ✓ Collision frequency; λ vs P and T — p18
- ✓ Moving container stopped; mixing; dissociation — p21, p22
- ✓ Standard results table; common traps — p23, p24
- ✓ Solved questions (8 Main + 6 Advanced level) — p25–p29

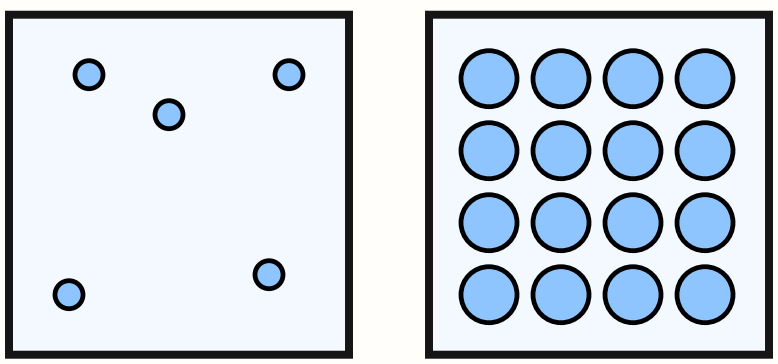
Ideal Gas: Assumptions

valid for a gas at LOW pressure and HIGH temperature (far from liquefaction)

1 Postulates of the kinetic theory

- **Size:** *molecules are point masses; their total volume is negligible compared with the container volume.*
- **Motion:** *random, in all directions with all speeds; on average no direction is preferred (isotropy, uniform density when gravity is neglected).*
- **Forces:** *no intermolecular force EXCEPT during a collision, so no intermolecular potential energy.*
- **Collisions:** *perfectly ELASTIC with each other and the walls; duration negligible compared with the time between collisions.*
- **Mechanics:** *Newton's laws apply to every molecule; the number of molecules is very large (statistics valid).*

2 Which assumption fails in a real gas?



low P: ideal *high P: real*

- *At HIGH pressure the molecules' own volume is not negligible.*
- *At LOW temperature attraction matters (slow molecules), so the gas can liquefy.*
- *Both effects vanish as $P \to 0$: every real gas becomes ideal.*

Ideal Gas Equation

ideal gas · thermal equilibrium · T in KELVIN · P in Pa, V in m^3

1 Equation of state

$$PV = nRT = NkT$$

$$P = n_V kT \qquad P = \frac{\rho RT}{M}$$

n = moles, N = molecules, $n_V = N/V$, ρ = density, M = molar mass · $k = \frac{R}{N_A} = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$ · IDEAL gas, any fixed sample in equilibrium

- For a FIXED amount (n constant): $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$ (the combined gas law).
- At the same P and T, all ideal gases have the same NUMBER density $n_V = \frac{P}{kT}$ (Avogadro's law: equal volumes hold equal N).

2 Standard numbers

Example · STP

STP: $T = 273.15\,\mathrm{K}$, $P = 1.013 \times 10^5\,\mathrm{Pa}$, $n = 1\,\mathrm{mol}$ of an ideal gas:
$V = \frac{RT}{P} = \frac{8.314 \times 273.15}{1.013 \times 10^5} = 0.0224\,\mathrm{m^3} = 22.4\,\mathrm{L}$.
Number density at STP:
$n_V = \frac{P}{kT} = \frac{1.013 \times 10^5}{1.38 \times 10^{-23} \times 273.15} = 2.69 \times 10^{25}\,\mathrm{m^{-3}}$.

Golden points

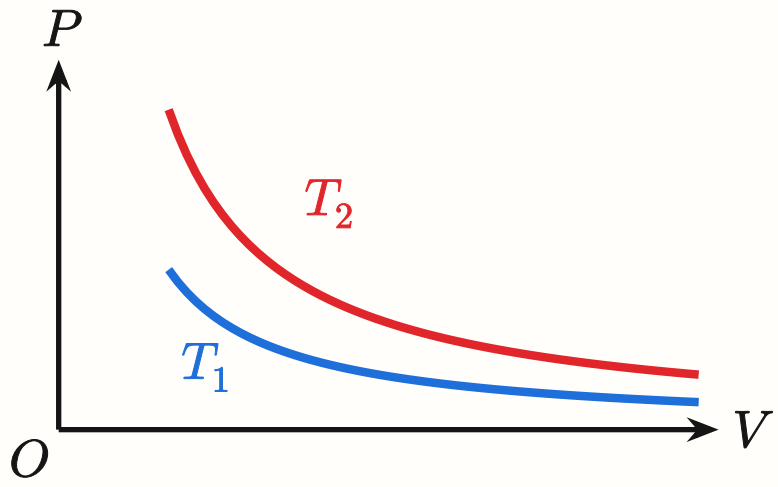
★ T in kelvin: $T(\mathrm{K}) = t(^\circ\mathrm{C}) + 273.15$. Never put °C into $PV = nRT$.

★ $1\,\mathrm{atm} = 1.013 \times 10^5\,\mathrm{Pa}$, $1\,\mathrm{L} = 10^{-3}\,\mathrm{m^3}$, $R = 0.0821\,\mathrm{L\,atm\,mol^{-1}K^{-1}}$.

Gas Laws

ideal gas · FIXED amount (n constant) · T in kelvin

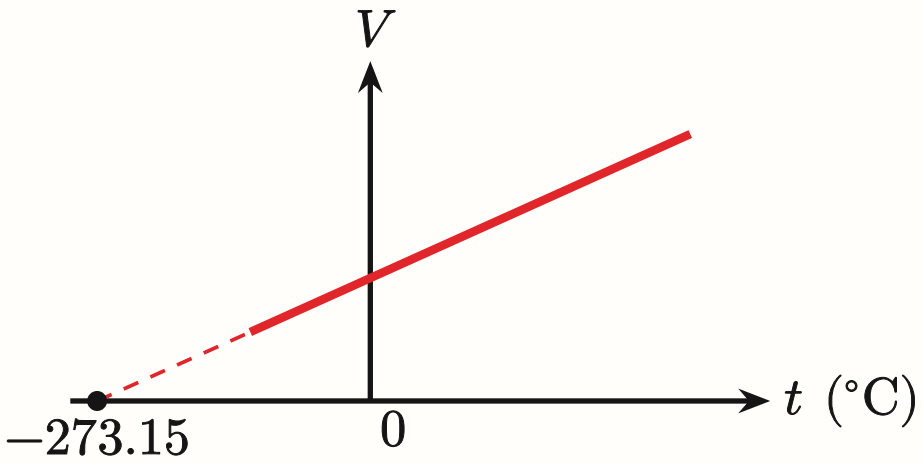
1 Boyle's law (T fixed)



$$PV = \text{constant}$$

n, T FIXED · isotherms: hyperbolas · $T_2 > T_1$: hotter isotherm FARTHER from the axes

2 Charles (P fixed), Gay-Lussac (V fixed)



$$V \propto T \qquad P \propto T$$

$V \propto T$: n, P fixed · $P \propto T$: n, V fixed · T in KELVIN · dashed: extrapolation to $-273.15\,°C$ (a real gas liquefies first)

3 Work done ON a gas

$$W_{on} = -\int_{V_1}^{V_2} P\,dV$$

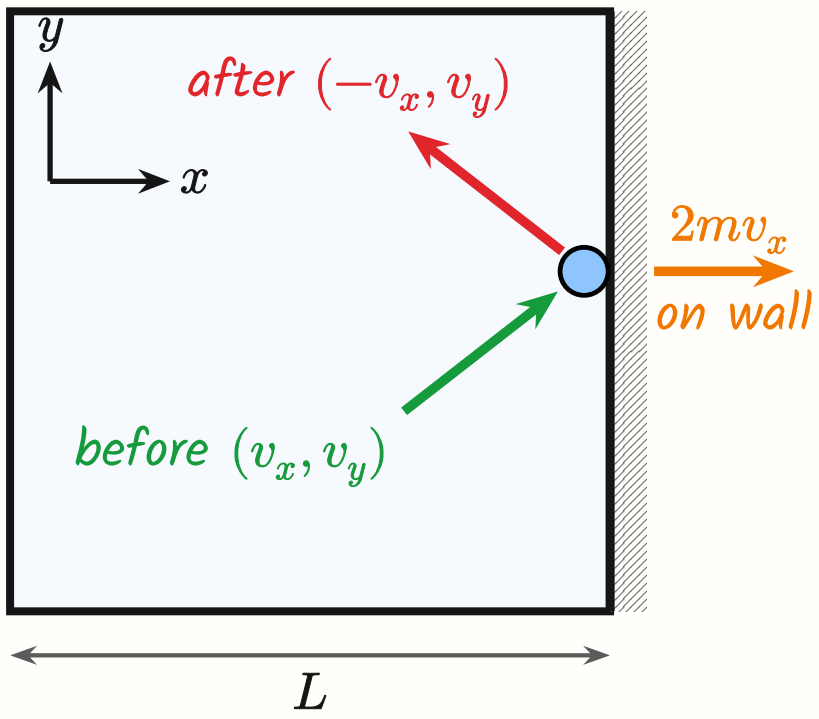
$$W_{on} = nRT \ln \frac{V_1}{V_2} \qquad W_{on} = -P\Delta V$$

QUASI-STATIC process $V_1 \to V_2$ · ISOTHERMAL ideal gas, fixed n · ISOBARIC · compression: $W_{on} > 0$

Pressure: Derivation

ideal gas · ELASTIC collisions with smooth walls · cube of side L · gravity neglected

1 One molecule hits the wall $x = L$



$$\Delta p_{wall} = 2mv_x$$

elastic, smooth wall: $v_x \to -v_x$, v_y, v_z unchanged · molecule's momentum change $-2mv_x$

- *Time between two hits on the SAME wall:* $\Delta t = \dfrac{2L}{v_x}$.
- *Average force by one molecule:* $\dfrac{2mv_x}{2L/v_x} = \dfrac{mv_x^2}{L}$.

2 Add all N molecules

$$F = \frac{m}{L}\sum v_x^2 = \frac{mN}{L}\langle v_x^2\rangle \qquad P = \frac{F}{L^2} = \frac{mN}{V}\langle v_x^2\rangle$$

$$\langle v_x^2\rangle = \langle v_y^2\rangle = \langle v_z^2\rangle = \frac{1}{3}\langle v^2\rangle \;\Rightarrow\; P = \frac{1}{3}\frac{N}{V}m\langle v^2\rangle$$

$\langle\ \rangle$ = average over molecules · isotropy: no preferred direction · $V = L^3$

- *Collisions BETWEEN molecules do not change the result: elastic collisions only exchange momentum, the averages stay the same.*
- *The result holds for ANY container shape (pressure is the same on every wall in equilibrium).*

Pressure: Results

ideal gas · thermal equilibrium · $v_{rms} = \sqrt{\langle v^2 \rangle}$

1 Kinetic interpretation of pressure

$$P = \frac{1}{3}\rho\, v_{rms}^2 = \frac{1}{3}\frac{N}{V} m\, v_{rms}^2$$

$$P = \frac{2}{3}\frac{E}{V} \qquad E = \frac{1}{2} N m v_{rms}^2$$

$\rho = \frac{Nm}{V}$ = density · E = total TRANSLATIONAL KE of the gas · $\frac{E}{V}$ in $\mathrm{J\,m^{-3}}$ = Pa

- Pressure = $\frac{2}{3}$ of the translational KE per unit volume (for ANY ideal gas, monatomic or not).
- At fixed ρ: $P \propto v_{rms}^2$. Doubling v_{rms} at fixed V and N makes P four times.

2 Worked example

Example

Ideal gas at $P = 1.0 \times 10^5\,\mathrm{Pa}$ with density $\rho = 1.2\,\mathrm{kg\,m^{-3}}$: $v_{rms} = \sqrt{\frac{3\times1.0\times10^5}{1.2}} = \sqrt{2.5\times10^5} = 500\,\mathrm{m\,s^{-1}}$; translational KE per unit volume $\frac{E}{V} = \frac{3}{2}P = 1.5\times10^5\,\mathrm{J\,m^{-3}}$.

Golden points

★ $v_{rms} = \sqrt{\langle v^2 \rangle} \geq \langle v \rangle$. Speeds $1, 2, 3, 4, 5\,\mathrm{m\,s^{-1}}$: mean 3, $v_{rms} = \sqrt{11} = 3.32\,\mathrm{m\,s^{-1}}$.

Kinetic Interpretation of Temperature

ideal gas · THERMAL EQUILIBRIUM · T in KELVIN · classical (not near absolute zero)

1 Average translational KE

$$\bar{E}_{tr} = \frac{1}{2}mv_{rms}^2 = \frac{3}{2}kT$$

per mole: $\frac{3}{2}RT$ $\qquad PV = \frac{2}{3}N\bar{E}_{tr}$

$\bar{E}_{tr}$ = average TRANSLATIONAL KE of ONE molecule · from $PV = \frac{1}{3}Nmv_{rms}^2$ and $PV = NkT$

- Temperature is a measure of the average translational KE of RANDOM motion. Bulk (ordered) motion of the container does not raise T.
- At the same T, EVERY ideal gas has the same $\bar{E}_{tr}$, whatever its mass or number of atoms.

2 Numbers at $300\,\mathrm{K}$

Example

$\bar{E}_{tr} = \frac{3}{2} \times 1.38 \times 10^{-23} \times 300 = 6.21 \times 10^{-21}\,\mathrm{J}$ per molecule $= 0.0388\,\mathrm{eV}$; per mole $\frac{3}{2} \times 8.314 \times 300 = 3741\,\mathrm{J}$.

Golden points

★ $\bar{E}_{tr} \propto T$ and $v_{rms} \propto \sqrt{T}$: doubling T (kelvin) doubles $\bar{E}_{tr}$ and multiplies v_{rms} by $\sqrt{2}$.

★ $27\,^\circ\mathrm{C}$ to $327\,^\circ\mathrm{C}$ is $300\,\mathrm{K} \to 600\,\mathrm{K}$: $\bar{E}_{tr}$ doubles (not $\times 12$).

Molecular Speeds

ideal gas obeying the MAXWELL distribution · thermal equilibrium · M in $\mathrm{kg\,mol^{-1}}$

1 Three speeds

$$v_{rms} = \sqrt{\frac{3RT}{M}} \qquad \bar{v} = \sqrt{\frac{8RT}{\pi M}} \qquad v_{mp} = \sqrt{\frac{2RT}{M}}$$

v_{rms} = root mean square, $\bar{v}$ = mean, v_{mp} = most probable speed · $\frac{R}{M} = \frac{k}{m}$, so $\sqrt{\frac{3kT}{m}}$ etc. · all $\propto \sqrt{T/M}$

$$v_{mp} : \bar{v} : v_{rms} = \sqrt{2} : \sqrt{\frac{8}{\pi}} : \sqrt{3} = 1 : 1.128 : 1.225$$

ALWAYS $v_{mp} < \bar{v} < v_{rms}$ (for the Maxwell distribution)

2 Oxygen at $300\,\mathrm{K}$

Example

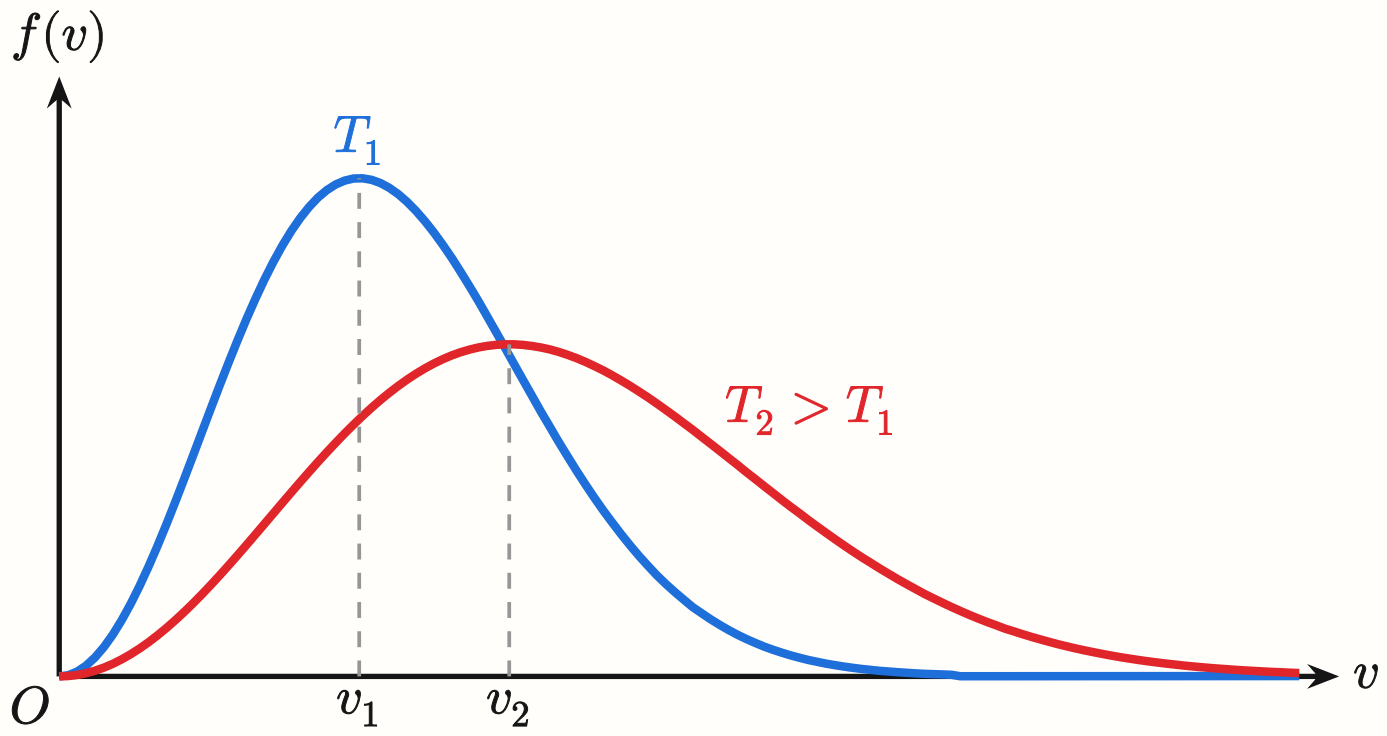
$M = 0.032\,\mathrm{kg\,mol^{-1}}$: $v_{rms} = \sqrt{\frac{3\times 8.314\times 300}{0.032}} = 483.6\,\mathrm{m\,s^{-1}}$, $\bar{v} = 445.6\,\mathrm{m\,s^{-1}}$, $v_{mp} = 394.8\,\mathrm{m\,s^{-1}}$.

- Same T, different gases: $\frac{v_1}{v_2} = \sqrt{\frac{M_2}{M_1}}$ (hydrogen vs oxygen: $\sqrt{32/2} = 4$).
- Same gas: $v \propto \sqrt{T}$. Speeds depend on T and M only: at fixed T a change of P changes ρ too, and $\sqrt{3P/\rho}$ stays the same.

Maxwell Speed Distribution

ideal gas · thermal equilibrium · SAME gas at two temperatures · curves normalised (area $= 1$)

1 Two temperatures, $T_2 > T_1$



v_1, v_2 = most probable speeds at T_1, T_2 (here $T_2 = 2.25T_1$, so $v_2 = 1.5v_1$).
Both curves start at the origin: $f(0) = 0$, almost no molecule is nearly at rest.

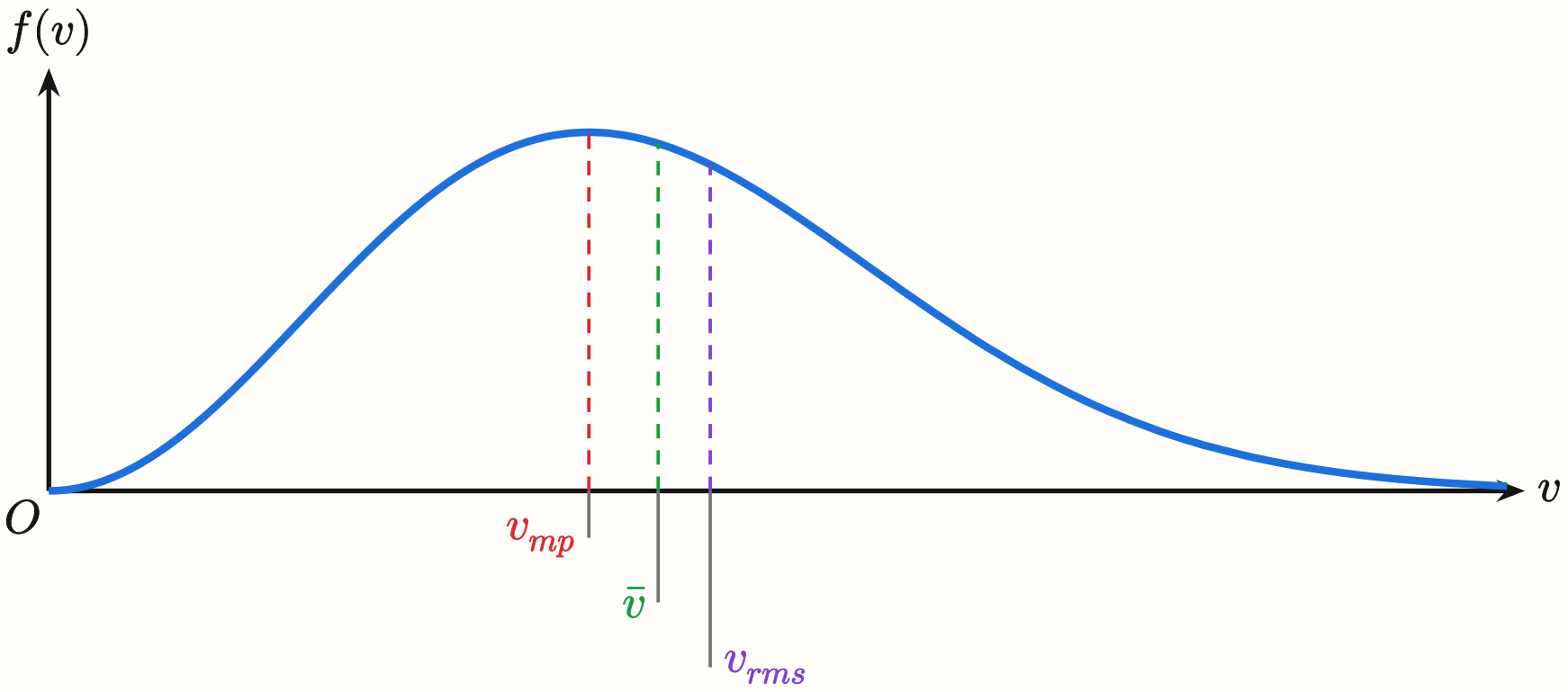
2 Graph features (qualitative)

- Hotter gas: the peak moves RIGHT ($v_{mp} \propto \sqrt{T}$) and becomes LOWER (height $\propto 1/v_{mp}$); the curve spreads out.
- AREA under each curve is the SAME ($= 1$, or N for dN/dv): the number of molecules is fixed.
- Fraction of molecules ABOVE a fixed high speed INCREASES with T (evaporation, escape from planets).

Speeds on the Curve

Maxwell distribution · ideal gas · thermal equilibrium

1 Where v_{mp}, $\bar{v}$, v_{rms} lie



Drawn to scale: $\bar{v} = 1.128\, v_{mp}$, $v_{rms} = 1.225\, v_{mp}$.

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2 Distribution function

$$f(v) = 4\pi \left(\frac{m}{2\pi kT}\right)^{3/2} v^2 e^{-mv^2/2kT} \qquad \int_0^\infty f\, dv = 1$$

$f(v)\,dv$ = *FRACTION of molecules with speed between* v *and* $v + dv$ ·
$dN = N f(v)\,dv$ · m = *mass of one molecule*

Golden points

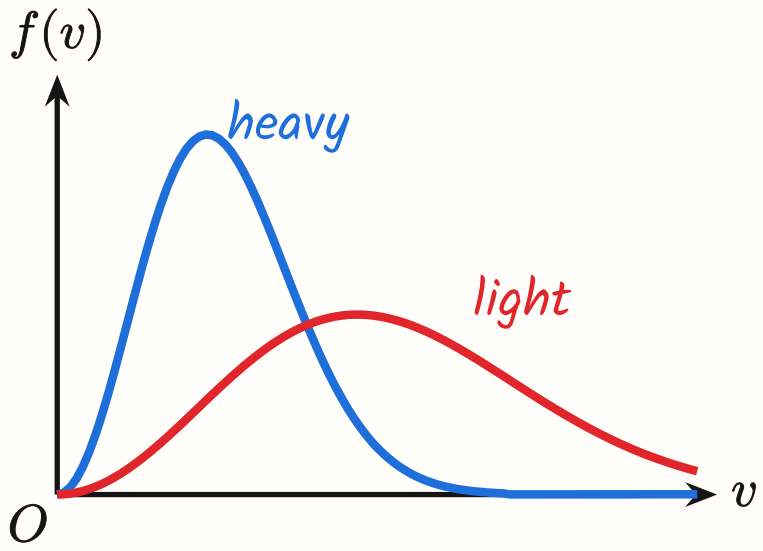
★ *Not symmetric: long tail to high speeds, so* $\bar{v}$ *and* v_{rms} *lie to the RIGHT of the peak.*

★ v_{mp}: *PEAK* $\left(\frac{df}{dv} = 0\right)$; $\bar{v} = \int_0^\infty v f\, dv$; $v_{rms}^2 = \int_0^\infty v^2 f\, dv$.

Maxwell: Effect of Mass

Maxwell distribution · ideal gases · thermal equilibrium · curves normalised (area $= 1$)

1 Effect of molar mass at the same T



- Lighter gas (smaller M) at the same T: curve like a HOTTER gas, peak right and lower.
- Here $M_{light} = \frac{1}{4} M_{heavy}$: v_{mp} doubles, peak height halves.

2 Peak position and height

$$v_{mp} = \sqrt{\frac{2kT}{m}} \qquad f_{max} = \frac{4}{\sqrt{\pi}\, e} \frac{1}{v_{mp}}$$

peak height $\propto \frac{1}{v_{mp}} \propto \sqrt{\frac{M}{T}}$ · e = base of natural logarithms · area under the curve fixed at 1

3 Same T, two gases

Example

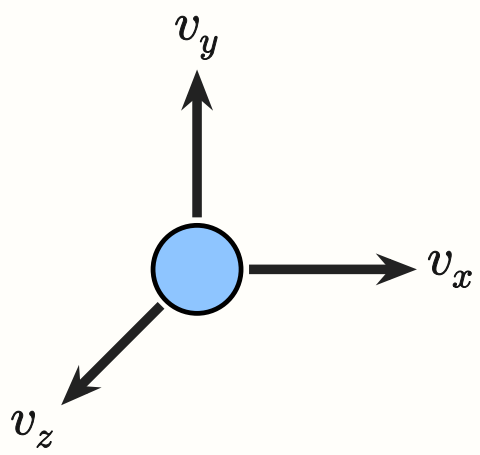
Hydrogen ($2\,\mathrm{g\,mol^{-1}}$) *and nitrogen* ($28\,\mathrm{g\,mol^{-1}}$) *at the same* T: $\frac{v_{mp}(\mathrm{H_2})}{v_{mp}(\mathrm{N_2})} = \sqrt{14} = 3.74$; *the hydrogen peak is* 3.74 *times LOWER.*

- Plotting $v_{mp} f(v)$ against v / v_{mp} gives ONE curve for every gas at every T.

Degrees of Freedom

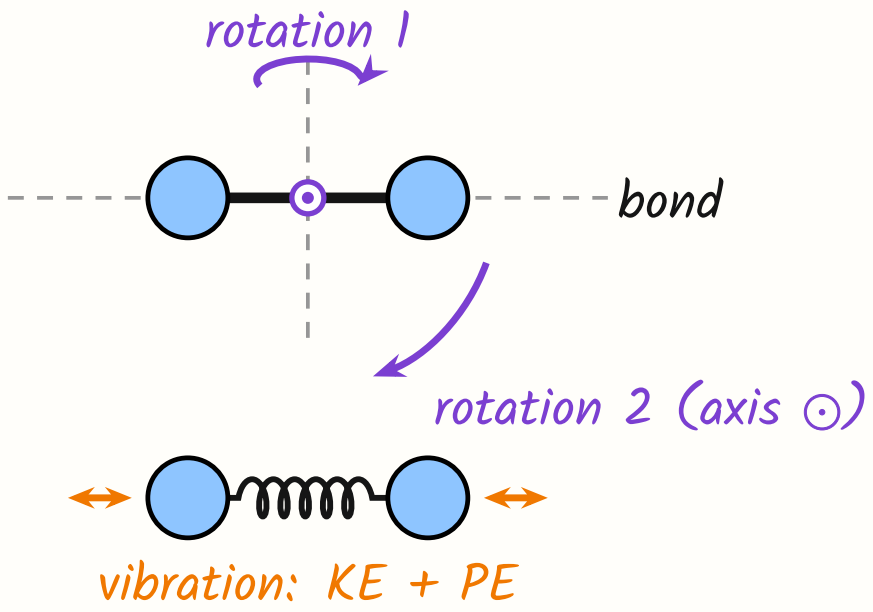
f = independent quadratic energy terms of ONE molecule

1 Monatomic: $f = 3$



- He, Ne, Ar: 3 translational only.

2 Diatomic: $f = 5$ rigid, $f = 7$ with vibration



- RIGID (H_2, N_2, O_2 at room T): 3 trans + 2 rot $= 5$ (bond-axis rotation: negligible moment of inertia).
- With VIBRATION (high T): $+2$ (KE + PE) $= 7$.

3 Polyatomic

- LINEAR rigid: $f = 5$. NON-LINEAR rigid (H_2O, NH_3, CH_4): 3 trans + 3 rot $= 6$.
- Each vibrational MODE adds 2 when excited. Frozen at room T for light molecules (H_2, N_2, O_2); CO_2 is treated as rigid ($f = 5$) only when the question says so (real CO_2 at $300\,K$: $\gamma \approx 1.29$).

Equipartition of Energy

classical ideal gas in THERMAL EQUILIBRIUM · T in kelvin · f = degrees of freedom

1 Law of equipartition

each quadratic term: $\frac{1}{2}kT$ per molecule

$$\bar{E} = \frac{f}{2}kT \text{ per molecule} \qquad \frac{f}{2}RT \text{ per mole}$$

every independent quadratic term ($\frac{1}{2}mv_x^2$, $\frac{1}{2}I\omega^2$, $\frac{1}{2}kx^2$ of a vibration) gets $\frac{1}{2}kT$ on average · a vibrational mode gets kT (KE + PE)

2 Internal energy of an ideal gas

$$U = \frac{f}{2}nRT = \frac{f}{2}NkT$$

ideal gas: no INTERmolecular PE, so U = KE of the molecules (plus vibrational PE when vibration is excited); U depends on T only · n = moles

- $\Delta U = \frac{f}{2}nR\,\Delta T$ for ANY process of an ideal gas with n and f fixed.
- Translational part is always $\frac{3}{2}nRT$; rotation and vibration add the rest.

Example

$2\,\text{mol}$ of a RIGID diatomic gas ($f = 5$) at $300\,\text{K}$:
$U = \frac{5}{2} \times 2 \times 8.314 \times 300 = 12471\,\text{J} \approx 12.5\,\text{kJ}$; its translational part $\frac{3}{2} \times 2 \times 8.314 \times 300 = 7483\,\text{J}$.

Specific Heats of Gases

IDEAL gas · MOLAR heat capacities · f fixed (no change in the excited modes)

1 From equipartition

$$C_V = \frac{f}{2}R \qquad C_P = C_V + R \qquad \gamma = \frac{C_P}{C_V} = 1 + \frac{2}{f}$$

C_V, C_P in $\mathrm{J\,mol^{-1}K^{-1}}$ · $C_P - C_V = R$ (Mayer): IDEAL gas only · per kg: $c_P - c_V = \frac{R}{M}$

- Mayer: at constant P, $Q = nC_V\Delta T + P\Delta V$ and $P\Delta V = nR\Delta T$, so $C_P = C_V + R$.

2 Table

Gas (condition)	f	C_V	C_P	γ
Monatomic	3	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3} = 1.67$
Diatomic, rigid	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5} = 1.4$
Diatomic, with vibration	7	$\frac{7}{2}R$	$\frac{9}{2}R$	$\frac{9}{7} = 1.29$
Linear polyatomic, rigid	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5} = 1.4$
Non-linear polyatomic, rigid	6	$3R$	$4R$	$\frac{4}{3} = 1.33$

Golden points

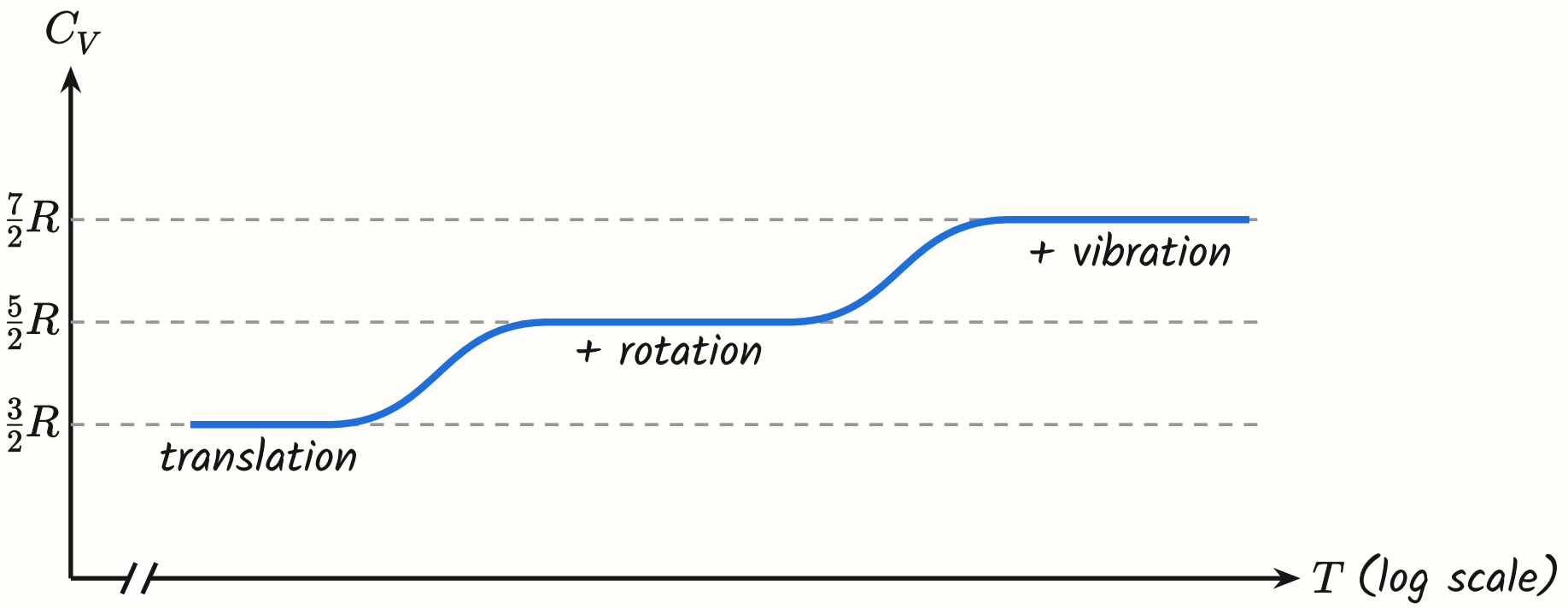
★ γ always lies between 1 and $\frac{5}{3}$ for an ideal gas; more degrees of freedom, smaller γ.

★ $f = \dfrac{2}{\gamma - 1}$ and $C_V = \dfrac{R}{\gamma - 1}$, $C_P = \dfrac{\gamma R}{\gamma - 1}$.

Heat Capacity and Temperature

qualitative graph · constant volume · gas treated as ideal · solids: per mole of atoms

1 C_V of hydrogen gas vs T



Axis break: the plot does not start at $T = 0$. Modes switch on only when kT is large enough (quantum effect). Room temperature: rotation active, vibration FROZEN, $f = 5$.

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2 Solids (Dulong–Petit)

$$C = 3R \approx 24.9\,\mathrm{J\,mol^{-1}K^{-1}}$$

solid: each atom vibrates in 3 directions, 3 KE + 3 PE terms, $\bar{E} = 3kT$ · Dulong–Petit, valid at HIGH (room) temperature, not for very light/hard solids (carbon) at room T

- *For a solid ΔV is tiny, so $C_P \approx C_V$; the Mayer relation is for ideal gases only.*
- *At low T the heat capacity of a solid falls towards 0: classical equipartition fails there.*

Mixture of Gases

ideal, NON-REACTING gases in thermal equilibrium · n_1, n_2 = moles · rigid molecules unless stated

1 Heat capacities and γ of a mixture

$$C_{V,mix} = \frac{n_1 C_{V1} + n_2 C_{V2}}{n_1 + n_2} \qquad C_{P,mix} = C_{V,mix} + R$$

$$\gamma_{mix} = \frac{C_{P,mix}}{C_{V,mix}} \qquad \frac{n_1 + n_2}{\gamma_{mix} - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1}$$

molar averages weighted by MOLES (not by mass) · never average the γ values directly

$$M_{mix} = \frac{n_1 M_1 + n_2 M_2}{n_1 + n_2} \qquad U = \frac{f_1}{2} n_1 RT + \frac{f_2}{2} n_2 RT$$

M = molar mass · common temperature T · f_1, f_2 = degrees of freedom

2 Worked example

Example

1 mol *helium* + 1 mol *oxygen (rigid)*: $C_V = \frac{\frac{3}{2}R + \frac{5}{2}R}{2} = 2R$, $C_P = 3R$, $\gamma = \frac{3}{2} = 1.5$. *Check:* $\frac{2}{\gamma - 1} = \frac{1}{2/3} + \frac{1}{2/5} = 4$, *so* $\gamma = 1.5$.
$M_{mix} = \frac{4 + 32}{2} = 18\,\mathrm{g\,mol^{-1}}$.

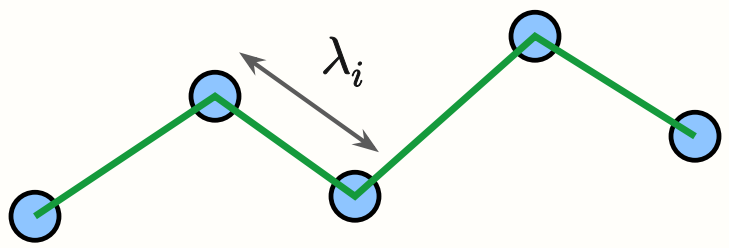
Golden points

★ *Mixing two ideal gases at the same T (insulated, rigid, non-reacting): T stays the same, U adds.*

Mean Free Path

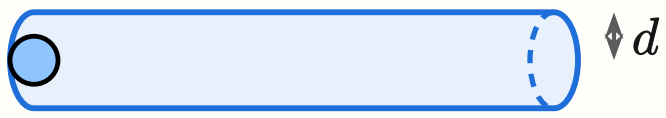
ideal gas of identical hard spheres of diameter d · Maxwell speeds · $n_V = N/V$ = number DENSITY

1 Free path between collisions



$$\lambda = \frac{1}{\sqrt{2}\,\pi d^2 n_V} = \frac{kT}{\sqrt{2}\,\pi d^2 P}$$

λ = MEAN distance between successive collisions (m) · πd^2 = collision cross-section · $\sqrt{2}$: relative motion of the other molecules



collision cylinder, radius d

- A molecule hits every molecule whose centre lies within d of its path: the swept cylinder of length $\bar{v}t$ holds $\pi d^2 \bar{v} t n_V$ molecules.
- Collision frequency $z = \dfrac{\bar{v}}{\lambda} = \sqrt{2}\,\pi d^2 n_V \bar{v}$; mean time between collisions $\frac{1}{z}$.

2 Dependence

- $\lambda \propto \frac{1}{n_V}$; FIXED V (sealed rigid box): λ does NOT change with T although P rises.
- At FIXED P: $\lambda \propto T$. At FIXED T: $\lambda \propto \frac{1}{P}$. λ does not depend on the molecular speed.

Example

Nitrogen at STP, $d = 3 \times 10^{-10}\,\mathrm{m}$, $n_V = 2.69 \times 10^{25}\,\mathrm{m}^{-3}$:
$\lambda = \dfrac{1}{\sqrt{2}\pi(3\times10^{-10})^2(2.69\times10^{25})} = 9.3 \times 10^{-8}\,\mathrm{m} \approx 310d$.

Dalton's Law and Graham's Law

ideal, NON-REACTING gases · same container (volume V) and same temperature T

1 Dalton's law of partial pressures

$$P = P_1 + P_2 + \dots \qquad P_i = \frac{n_i RT}{V} = x_i P$$

P_i = partial pressure: the pressure gas i would exert ALONE in the whole volume at the same T · $x_i = \frac{n_i}{n}$ = mole fraction

- Kinetic reason: each gas delivers its own momentum to the walls; no intermolecular forces, so the effects add.
- Fails for gases that react (e.g. $\mathrm{NH_3}$ + HCl) or at high pressure.

Example

$4\,\mathrm{g}$ hydrogen ($2\,\mathrm{mol}$) + $32\,\mathrm{g}$ oxygen ($1\,\mathrm{mol}$), total $P = 3 \times 10^5\,\mathrm{Pa}$: $x_{\mathrm{H_2}} = \frac{2}{3}$, $P_{\mathrm{H_2}} = 2 \times 10^5\,\mathrm{Pa}$, $P_{\mathrm{O_2}} = 1 \times 10^5\,\mathrm{Pa}$.

2 Graham's law of effusion / diffusion

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{\rho_2}{\rho_1}}$$

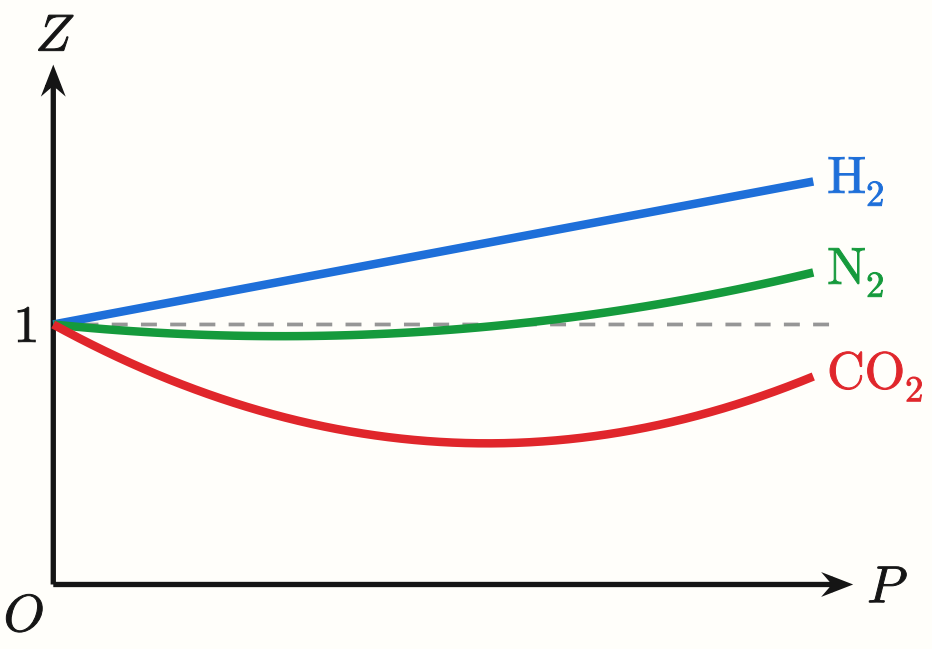
r = rate (molecules or volume per second) through a SMALL hole (smaller than λ) · same P and T · rate $\propto \bar{v} \propto \frac{1}{\sqrt{M}}$

- Hydrogen effuses $\sqrt{32/2} = 4$ times faster than oxygen.
- Used to separate isotopes: $^{235}\mathrm{UF_6}$ and $^{238}\mathrm{UF_6}$ differ by only 0.43% in rate per stage.

Real Gases

real gases · qualitative · all curves at the SAME temperature, above the critical temperature of each gas shown (e.g. $320\,\mathrm{K}$)

1 Compressibility factor Z



$$Z = \frac{PV}{nRT}$$

dashed line: ideal gas, $Z = 1$ at all P · every real gas: $Z \to 1$ as $P \to 0$

- $Z < 1$: *attraction dominates (easier to compress).*
- $Z > 1$: *molecular size dominates (high P).*

2 Van der Waals equation (qualitative)

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

a: attraction (lowers the wall pressure) · b: excluded volume per mole (molecules' own size) · $a = b = 0$ gives the ideal gas

- *Real gases behave IDEALLY at LOW pressure and HIGH temperature (molecules far apart, KE $\gg$ attraction).*
- *Deviation is large near liquefaction: high P, low T.*
- *Hydrogen and helium are closest to ideal at room temperature (weak attraction).*

Special Cases (I)

ideal gas · conditions stated in each case

1 Moving container stopped suddenly

$$\frac{1}{2}nMv_0^2 = nC_V\Delta T \;\Rightarrow\; \Delta T = \frac{Mv_0^2}{2C_V} = \frac{Mv_0^2}{fR}$$

INSULATED container, heat capacity of the walls neglected, all ordered KE becomes internal energy · v_0 = speed of the container · M in $\mathrm{kg\,mol^{-1}}$

- While the container moves uniformly, T does NOT change: ordered motion is not random motion.

2 Mixing two samples (insulated, rigid, no work)

$$T_f = \frac{n_1 f_1 T_1 + n_2 f_2 T_2}{n_1 f_1 + n_2 f_2}$$

total U conserved · same kind of gas ($f_1 = f_2$): $T_f = \frac{n_1 T_1 + n_2 T_2}{n_1 + n_2}$

3 Scaling at a glance

- v_{rms} doubled in a rigid sealed vessel: $T \times 4$, $P \times 4$, $\bar{E}_{tr} \times 4$.
- Isothermal compression: v_{rms}, $\bar{E}_{tr}$ unchanged; P rises because n_V rises (more hits per second).
- Half the gas leaks out at constant V and T: P halves, v_{rms} unchanged, λ doubles.
- Lighter gases ($\mathrm{H_2}$, He) are rare in the earth's atmosphere: a larger fraction of their molecules exceeds the escape speed $11.2\,\mathrm{km\,s^{-1}}$.

Special Cases (2)

ideal gas · conditions stated in each case

1 Dissociation of a diatomic gas

- $X_2 \to 2X$: the number of moles INCREASES ($n \to n(1+\alpha)$ for a fraction α dissociated), so at fixed V and T the pressure rises.
- The atoms are monatomic ($f = 3$), the remaining molecules diatomic: use $U = \sum \frac{f_i}{2} n_i RT$ (dissociation energy not included).

2 Two vessels joined by a thin tube

$$\frac{P_1 V_1}{T_1} + \frac{P_2 V_2}{T_2} = P\left(\frac{V_1}{T_1'} + \frac{V_2}{T_2'}\right)$$

total MOLES conserved (no leak) · common final pressure P · T_1', T_2' = final temperatures of the two vessels (each vessel held at its own temperature)

3 Average values

- Average VELOCITY of the molecules $= \vec{0}$ (gas with no bulk motion, container at rest); average SPEED $\bar{v} \neq 0$.
- $\langle v_x \rangle = 0$ but $\langle v_x^2 \rangle = \frac{1}{3} v_{rms}^2$; average KE along one axis $= \frac{1}{2} kT$.
- Each wall collision delivers $2m|v_x| > 0$, so the pressure is not zero even though $\langle v_x \rangle = 0$.

Standard Results

1 One-line results

Quantity	Result	Condition
Gas equation	$PV = nRT = NkT$	ideal gas, T in K
Pressure	$P = \frac{1}{3}\rho v_{rms}^2 = \frac{2}{3}\frac{E}{V}$	ideal gas, E translational
KE per molecule	$\frac{1}{2}mv_{rms}^2 = \frac{3}{2}kT$	translational, equilibrium
rms speed	$\sqrt{\frac{3RT}{M}} = \sqrt{\frac{3P}{\rho}}$	M in $\mathrm{kg\,mol^{-1}}$
Mean speed	$\sqrt{\frac{8RT}{\pi M}}$	Maxwell distribution
Most probable	$\sqrt{\frac{2RT}{M}}$	Maxwell distribution
Ratio	$1 : 1.128 : 1.225$	$v_{mp} : \bar{v} : v_{rms}$
Energy per mole	$\frac{f}{2}RT$	equipartition
Internal energy	$U = \frac{f}{2}nRT$	ideal gas
Heat capacities	$C_V = \frac{f}{2}R,\ C_P = C_V + R$	ideal gas, molar
γ	$1 + \frac{2}{f}$	f fixed
Mixture	$C_V = \frac{\sum n_i C_{Vi}}{\sum n_i}$	non-reacting
Mean free path	$\frac{1}{\sqrt{2}\pi d^2 n_V}$	hard spheres
Dalton	$P = \sum P_i$	non-reacting, ideal
Graham	$r \propto \frac{1}{\sqrt{M}}$	same P, T
Solids	$C = 3R$	high T (Dulong–Petit)

Summary

$PV = nRT$ • $\frac{1}{2}mv_{rms}^2 = \frac{3}{2}kT$ • $U = \frac{f}{2}nRT$ • $\gamma = 1 + \frac{2}{f}$

ideal gas · thermal equilibrium · T in kelvin · f fixed

Common Traps

1 Exam traps

Avoid these

- ✗ Using °C in $PV = nRT$ or $v_{rms} \propto \sqrt{T}$ (no: kelvin).
- ✗ Using $M = 32$ for oxygen in $\sqrt{3RT/M}$ (no: $0.032\,\mathrm{kg\,mol^{-1}}$).
- ✗ Average KE $\frac{3}{2}kT$ is the TOTAL KE of a diatomic molecule (no: translational only; total $\frac{5}{2}kT$ if rigid).
- ✗ v_{rms} = average speed (no: $v_{rms} > \bar{v} > v_{mp}$).
- ✗ Mean free path rises with T in a sealed rigid vessel (no: n_V fixed, λ fixed).
- ✗ Averaging γ of a mixture directly (no: average C_V by moles).
- ✗ Weighting molar C_V of a mixture by MASS (no: by moles).
- ✗ $C_P - C_V = R$ for solids and liquids (no: ideal gases only).
- ✗ Higher T makes the Maxwell peak taller (no: lower and to the right).
- ✗ A moving container's gas is hotter (no: only random motion counts).
- ✗ Vibration is counted at room temperature (no: frozen for $\mathrm{H_2}$, $\mathrm{N_2}$, $\mathrm{O_2}$ at room T, $f = 5$).
- ✗ $\frac{3}{2}kT$ per mole (no: per molecule; $\frac{3}{2}RT$ per mole).

Golden points

- ★ First ask: per molecule or per mole? translational or total? kelvin? moles or mass? rigid or vibrating?

Questions: JEE Main Level (I)

ideal gases in equilibrium · T in K · $R = 8.314$, $k = 1.38 \times 10^{-23}$ (SI)

Q1 · JEE Main level · MCQ

The rms speed of the molecules of an ideal gas at $27\,^\circ\mathrm{C}$ is v. At what temperature is it $2v$? (A) $108\,^\circ\mathrm{C}$ (B) $927\,^\circ\mathrm{C}$ (C) $1200\,^\circ\mathrm{C}$ (D) $54\,^\circ\mathrm{C}$

Solution: $v_{rms} \propto \sqrt{T}$: $T = 4 \times 300 = 1200\,\mathrm{K} = 927\,^\circ\mathrm{C}$.

Answer: (B)

Q2 · JEE Main level · Numerical

An ideal gas at pressure $2.0 \times 10^5\,\mathrm{Pa}$ has density $1.5\,\mathrm{kg\,m^{-3}}$. Find v_{rms} (in $\mathrm{m\,s^{-1}}$).

Solution: $v_{rms} = \sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3\times 2.0\times 10^5}{1.5}} = \sqrt{4\times 10^5} = 632.5\,\mathrm{m\,s^{-1}}$.

Answer: 632.5

Q3 · JEE Main level · Numerical

$2\,\mathrm{mol}$ monatomic + $3\,\mathrm{mol}$ rigid diatomic ideal gas (non-reacting). γ of the mixture?

Solution: $C_V = \frac{2\times\frac{3}{2}R + 3\times\frac{5}{2}R}{5} = 2.1R$, $C_P = 3.1R$, $\gamma = \frac{3.1}{2.1} = 1.476$.

Answer: 1.48

Q4 · JEE Main level · MCQ

Helium ($4\,\mathrm{g\,mol^{-1}}$) and methane ($16\,\mathrm{g\,mol^{-1}}$) at the same T. Ratios of v_{rms} ($\mathrm{He} : \mathrm{CH_4}$) and of average translational KE per molecule: (A) $2:1$, $1:1$ (B) $1:2$, $1:1$ (C) $2:1$, $4:1$ (D) $4:1$, $1:1$

Solution: $v_{rms} \propto \frac{1}{\sqrt{M}}$; $\sqrt{16/4} = 2$; $\bar{E}_{tr} = \frac{3}{2}kT$ is the same.

Answer: (A)

Questions: JEE Main Level (2)

ideal gases in equilibrium · T in K · $R = 8.314$, $k = 1.38 \times 10^{-23}$ (SI)

Q5 · JEE Main level · MCQ

At constant temperature the pressure of a fixed amount of an ideal gas is made 3 times. The mean free path becomes: (A) 3λ (B) $\frac{\lambda}{3}$ (C) λ (D) $\frac{\lambda}{\sqrt{3}}$

Solution: $\lambda = \frac{kT}{\sqrt{2}\pi d^2 P} \propto \frac{1}{P}$ at fixed T (same d).

Answer: (B)

Q6 · JEE Main level · Numerical

Find the internal energy of $3\,\text{mol}$ of a monatomic ideal gas at $400\,\text{K}$ (in kJ).

Solution: $U = \frac{3}{2}nRT = \frac{3}{2} \times 3 \times 8.314 \times 400 = 14965\,\text{J}$.

Answer: $14.97\,\text{kJ}$

Q7 · JEE Main level · Numerical

A $10\,\text{L}$ vessel at $300\,\text{K}$ holds $0.2\,\text{mol}$ nitrogen and $0.3\,\text{mol}$ oxygen (ideal, non-reacting). Total pressure and partial pressure of oxygen?

Solution: $P = \frac{0.5 \times 8.314 \times 300}{0.010} = 1.247 \times 10^5\,\text{Pa}$;
$P_{\text{O}_2} = 0.6P = 7.48 \times 10^4\,\text{Pa}$.

Answer: $1.25 \times 10^5\,\text{Pa}$, $7.48 \times 10^4\,\text{Pa}$

Q8 · JEE Main level · Numerical

How many molecules are in $1\,\text{cm}^3$ of an ideal gas at $27\,^\circ\text{C}$ and $1.38 \times 10^5\,\text{Pa}$?

Solution: $N = \frac{PV}{kT} = \frac{1.38 \times 10^5 \times 10^{-6}}{1.38 \times 10^{-23} \times 300} = 3.33 \times 10^{19}$.

Answer: 3.33×10^{19}

Questions: JEE Advanced Level (I)

ideal gases in equilibrium · T in K · $R = 8.314$, $k = 1.38 \times 10^{-23}$ (SI)

A1 · JEE Advanced level · Numerical, multi-step

A thermally insulated rigid vessel containing nitrogen (rigid diatomic, $M = 0.028\,\mathrm{kg\,mol^{-1}}$) at $300\,\mathrm{K}$ moves at $100\,\mathrm{m\,s^{-1}}$ and is stopped suddenly. The walls take no heat. Find (i) the rise in temperature, (ii) the percentage rise in pressure, (iii) the new v_{rms}.

Solution: (i) $\frac{1}{2}Mv_0^2 = C_V\Delta T$ per mole: $\Delta T = \frac{0.028\times100^2}{5\times8.314} = 6.74\,\mathrm{K}$. (ii) Fixed V, n: $\frac{\Delta P}{P} = \frac{\Delta T}{T} = \frac{6.74}{300} = 2.25\%$. (iii) $v_{rms} = \sqrt{\frac{3\times8.314\times306.74}{0.028}} = 522.7\,\mathrm{m\,s^{-1}}$ (was 516.9).

Answer: (i) $6.74\,\mathrm{K}$ (ii) 2.25% (iii) $522.7\,\mathrm{m\,s^{-1}}$

A2 · JEE Advanced level · Numerical, multi-step

An insulated rigid vessel has two parts: $2\,\mathrm{mol}$ helium at $300\,\mathrm{K}$ and $1\,\mathrm{mol}$ oxygen (rigid) at $600\,\mathrm{K}$. The partition is removed (no reaction). Find (i) the final temperature, (ii) γ of the mixture, (iii) $v_{rms}(\mathrm{He})/v_{rms}(\mathrm{O_2})$ in the final state.

Solution: (i) U conserved: $T_f = \frac{2\times\frac{3}{2}\times300+1\times\frac{5}{2}\times600}{2\times\frac{3}{2}+1\times\frac{5}{2}} = \frac{2400}{5.5} = 436.4\,\mathrm{K}$. (ii) $C_V = \frac{5.5R}{3} = 1.833R$, $C_P = 2.833R$, $\gamma = \frac{17}{11} = 1.545$. (iii) Same T: $\sqrt{\frac{32}{4}} = 2.83$.

Answer: (i) $436.4\,\mathrm{K}$ (ii) 1.545 (iii) 2.83

Golden points

★ Stopped container: $\Delta T = \frac{Mv_0^2}{fR}$ does not depend on the amount of gas. Mixing: conserve $U = \sum \frac{f_i}{2}n_iRT$, not T.

Questions: JEE Advanced Level (2)

ideal gases in equilibrium · T in K · $R = 8.314$, $k = 1.38 \times 10^{-23}$ (SI)

A3 · JEE Advanced level · Multi-correct

A fixed amount of an ideal gas (Maxwell distribution) is heated from T to $4T$. Correct statements: (A) v_{mp} doubles (B) the peak height of $f(v)$ halves (C) the area under $f(v)$ doubles (D) the fraction of molecules slower than the OLD v_{mp} decreases

Solution: $v_{mp} \propto \sqrt{T}$: doubles. Peak $f_{max} = \frac{4}{\sqrt{\pi} e\, v_{mp}} \propto \frac{1}{v_{mp}}$: halves. Area $= 1$: (C) false. Fraction below v: $F(x) = \operatorname{erf}(x) - \frac{2}{\sqrt{\pi}} x e^{-x^2}$, $x = \frac{v}{v_{mp}}$: before $F(1) = 0.428$, after $F(0.5) = 0.081$: decreases.

Answer: (A), (B), (D)

A4 · JEE Advanced level · Numerical, multi-step

Helium ($M = 0.004\,\mathrm{kg\,mol^{-1}}$, molecular diameter $2.0 \times 10^{-10}\,\mathrm{m}$) is at $400\,\mathrm{K}$ and $2.0 \times 10^{5}\,\mathrm{Pa}$. Find (i) n_V, (ii) the mean free path, (iii) the mean speed, (iv) the collision frequency.

Solution: (i) $n_V = \frac{P}{kT} = \frac{2.0\times10^{5}}{1.38\times10^{-23}\times400} = 3.62 \times 10^{25}\,\mathrm{m^{-3}}$. (ii) $\lambda = \frac{1}{\sqrt{2}\pi(2.0\times10^{-10})^2\times3.62\times10^{25}} = 1.55 \times 10^{-7}\,\mathrm{m}$. (iii) $\bar{v} = \sqrt{\frac{8\times8.314\times400}{\pi\times0.004}} = 1455\,\mathrm{m\,s^{-1}}$. (iv) $z = \frac{\bar{v}}{\lambda} = 9.37 \times 10^{9}\,\mathrm{s^{-1}}$.

Answer: (i) $3.62 \times 10^{25}\,\mathrm{m^{-3}}$ **(ii)** $1.55 \times 10^{-7}\,\mathrm{m}$ **(iii)** $1455\,\mathrm{m\,s^{-1}}$ **(iv)** $9.37 \times 10^{9}\,\mathrm{s^{-1}}$

Golden points

★ $z \propto P/\sqrt{TM}$ at fixed d ($n_V = \frac{P}{kT}$, $\bar{v} \propto \sqrt{T/M}$).

Questions: JEE Advanced Level (3)

ideal gases in equilibrium · T in K · $R = 8.314$, $k = 1.38 \times 10^{-23}$ (SI)

AS · JEE Advanced level · Numerical, multi-step

A rigid sealed vessel holds a diatomic ideal gas at $300\,\mathrm{K}$ (rigid molecules). It is heated to $3000\,\mathrm{K}$, where 50% of the molecules dissociate into atoms. Treat the remaining molecules as rigid ($f = 5$, given) and leave out the dissociation energy. Find (i) P_f/P_i, (ii) U_f/U_i (thermal KE only).

Solution: (i) Moles: $n \to 0.5n + 2 \times 0.5n = 1.5n$.
$\frac{P_f}{P_i} = \frac{1.5 \times 3000}{300} = 15$. (ii) $U_i = \frac{5}{2} nR(300) = 750nR$;
$U_f = \left(\frac{5}{2} \times 0.5n + \frac{3}{2} \times n\right) R(3000) = 8250nR$; ratio 11.

Answer: (i) 15 (ii) 11

A6 · JEE Advanced level · Numerical, multi-step

EQUAL MASSES of helium ($4\,\mathrm{g\,mol^{-1}}$) and oxygen ($32\,\mathrm{g\,mol^{-1}}$, rigid) are mixed at the same temperature (ideal, non-reacting). Find (i) the mole ratio, (ii) γ of the mixture, (iii) the fraction of the internal energy carried by helium.

Solution: (i) $n_{\mathrm{He}} : n_{\mathrm{O_2}} = \frac{1}{4} : \frac{1}{32} = 8 : 1$. (ii)
$C_V = \frac{8 \times 1.5 + 1 \times 2.5}{9} R = 1.611R$, $C_P = 2.611R$, $\gamma = \frac{23.5}{14.5} = 1.621$. (iii)
$\frac{8 \times 1.5}{8 \times 1.5 + 2.5} = \frac{12}{14.5} = 0.828$.

Answer: (i) $8 : 1$ (ii) 1.621 (iii) 0.828

Golden points

★ Mixtures: always convert MASS to MOLES first, then weight C_V by moles.